

A Precomputation

During optimization, the tensor A (which stores the polynomial coefficients of each vertex function) is the only quantity that is updated. Hence, we obtain substantial speedups by precomputing quantities that remain fixed. Before the optimization loop, we generate N samples per triangle of the coarse mesh and store them as a $N \times F \times 2$ tensor $\mathbf{x}$. The 2D samples are drawn uniformly at random on the canonical equilateral triangle. On these samples, we precompute the one-ring coordinates, blend weights, polynomial basis (i.e., $\Phi(\text{oc}(\mathbf{x}))$), rotations, radii, and angles.

During training, we draw batches of points per face from this precomputed pool. This substantially reduces the time required to evaluate the Blended Chart Surface output points prior to updating the polynomial coefficients. If the Jacobian of the parametrization is needed (e.g., for area-weighted sampling, normal regularization, or distortion regularization), we additionally precompute gradients to reduce the amount of `autograd` computation required in the training loop. The Jacobian of the parametrization is,

$$\begin{aligned} \frac{\partial c}{\partial \mathbf{x}} &= \frac{\partial c}{\partial \text{oc}(\mathbf{x})} \boxed{\frac{d \text{oc}(\mathbf{x})}{d \mathbf{x}}} + \frac{\partial c}{\partial B^{2D}(\mathbf{x})} \boxed{\frac{d B^{2D}(\mathbf{x})}{d \mathbf{x}}} \\ &+ \frac{\partial c}{\partial \Phi} \boxed{\frac{d \Phi}{d \text{oc}(\mathbf{x})}} \boxed{\frac{d \text{oc}(\mathbf{x})}{d \mathbf{x}}} \end{aligned} \quad (\text{Supp.1})$$

All quantities should be understood as tensors stacked over the appropriate indices (face indices and/or per-face vertex indices). Note that the boxed terms do not depend on the polynomial coefficients and can therefore be precomputed. All boxed terms are obtained via `autograd` except for the polynomial-basis term $\frac{d\Phi}{d\text{oc}(\mathbf{x})}$. While this term could also be computed with `autograd`, it is simpler to exploit the closed-form differentiability of polynomials. If $\text{oc}(\mathbf{x}) = (u, v)$, then for quadratic vertex functions (as used in the surface examples),

$$\begin{aligned} \frac{d\Phi}{d\text{oc}(\mathbf{x})} &= \frac{d}{d(u, v)} \begin{bmatrix} 1 & u & v & u^2 & uv & v^2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 & 2u & v & 0 \\ 0 & 0 & 1 & 0 & u & 2v \end{bmatrix}. \end{aligned} \quad (\text{Supp.2})$$

B Proofs and Technical Details

B.1 Equivariance to Rotation and Scaling

For both Blended Chart Curves and Blended Chart Surfaces (which we will jointly refer to as Blended Chart Shapes), the local geometry is defined by vertex functions $p_i(t) = s_i R_i m_i(t) + v_i$, which are blended together via

$$c_{i,i+1}(t) = p_i(t) B(t) + p_{i+1}(\tau_{i,i+1}(t)) B(\tau_{i,i+1}(t)) \text{ for } t \in (0, 1]. \quad (\text{Supp.3})$$

(for curves) or

$$c_{ijk}(\mathbf{x}) = p_i(\text{oc}_i(\mathbf{x})) B^{2D}(\mathbf{x}, i) + p_j(\text{oc}_j(\mathbf{x})) B^{2D}(\mathbf{x}, j) + p_k(\text{oc}_k(\mathbf{x})) B^{2D}(\mathbf{x}, k) \quad (\text{Supp.4})$$

(for surfaces).

For the Blended Chart Shape to be equivariant to scaling, it means that if we scale the proxy (mesh) geometry by s , then the function c will also be scaled by a factor of s .

Suppose we scale the proxy mesh geometry by s . Then v_i scales by s , and s_i also scales by s because we defined the local scale s_i to be the mean of the outgoing edge-lengths. So clearly, p_i also scales by s . Then we see that every term in the definition of c_i scales by s . So the Blended Chart Shape is indeed equivariant to scaling.

Similarly, if we rotate the proxy geometry by R , then $v_i \mapsto R v_i$, and also $R_i \mapsto R R_i$ because R_i was defined to be a local coordinate frame based on the proxy mesh geometry. So overall, $p_i(t) \mapsto R p_i(t)$. Every term in the definition of c becomes rotated by R , therefore $c \mapsto R c$, and c is equivariant to rotations of the proxy mesh geometry, as we wanted.

The key part of the formulation that allows equivariance to scale and rotation is that the vertex functions p_i are constructed in a local coordinate frames R_i based on the proxy geometry, and p_i is proportional to a ‘local scale’ term s_i .

B.2 Equivariance to Translation, Partition of Unity

A Blended Chart Shape (curve or surface) is equivariant to translations of the coarse points if and only if the Blending Function satisfies the Partition of Unity (PoU) property.

Proof for Blended Chart Curves:

A blended chart curve is locally defined by the equation

$$c_{i,i+1}(t) = p_i(t)B(t) + p_{i+1}(\tau_{i,i+1}(t))B(\tau_{i,i+1}(t)) \text{ for } t \in (0, 1] \quad (\text{Supp.5})$$

with $p_i(t) = s_i R_i m_i(t) + v_i$.

Now let

$$\hat{c}_{i,i+1}(t) = \hat{p}_i(t)B(t) + \hat{p}_{i+1}(\tau_{i,i+1}(t))B(\tau_{i,i+1}(t)) \text{ for } t \in (0, 1] \quad (\text{Supp.6})$$

for $\hat{p}_i(t) = s_i R_i m_i(t) + (v_i + T)$ where T is some (nonzero) translation applied to all the coarse points. i.e., $\hat{p}_i(t) = p_i(t) + T$.

The blended chart curve is (locally) equivariant to translations of the coarse points $\Leftrightarrow$

$$\hat{c}_{i,i+1}(t) = c_{i,i+1}(t) + T \text{ for } t \in (0, 1]$$

$\Leftrightarrow$

$$\begin{aligned} & \hat{p}_i(t)B(t) + \hat{p}_{i+1}(\tau_{i,i+1}(t))B(\tau_{i,i+1}(t)) = \\ & p_i(t)B(t) + p_{i+1}(\tau_{i,i+1}(t))B(\tau_{i,i+1}(t)) + T \text{ for } t \in (0, 1] \end{aligned}$$

$\Leftrightarrow$

$$\begin{aligned} & (p_i(t) + T)B(t) + (p_{i+1}(\tau_{i,i+1}(t) + T))B(\tau_{i,i+1}(t)) = \\ & p_i(t)B(t) + p_{i+1}(\tau_{i,i+1}(t))B(\tau_{i,i+1}(t)) + T \text{ for } t \in (0, 1] \end{aligned}$$

$\Leftrightarrow$

$$B(t)T + B(\tau_{i,i+1}(t))T = T \text{ for } t \in [-1, 0]$$

(by subtracting the common terms on both sides),

Factoring out T , and using the assumption that $T \neq 0$, we see that the equation holds if and only if

$$B(t) + B(\tau_{i,i+1}(t)) = 1 \text{ for } t \in (0, 1]$$

This is exactly the PoU property. So, equivariance to translation holds if and only if the blending function has PoU.

The same argument may be followed to show that Blended Chart Surfaces are equivariant to translation if and only if the 2D Blending functions satisfy PoU.

Proof for Surfaces: The proof for Blended Chart Surfaces is completely analogous. Define $\hat{p}_i$ in the same way as above, for an arbitrary non-zero translation T .

The BCS is locally defined by

$$c_{ijk}(\mathbf{x}) = p_i(\text{oc}_i^{ijk}(\mathbf{x}))B^{2D}(\mathbf{x}, i) + p_j(\text{oc}_j^{ijk}(\mathbf{x}))B^{2D}(\mathbf{x}, j) + p_k(\text{oc}_k^{ijk}(\mathbf{x}))B^{2D}(\mathbf{x}, k) \quad (\text{Supp.6})$$

for $\mathbf{x}$ in the unit-length equilateral triangle which we will denote by E .

$$\begin{aligned} & \text{Let} \\ & \hat{c}_{ijk}(\mathbf{x}) = \hat{p}_i(\text{oc}_i^{ijk}(\mathbf{x}))B^{2D}(\mathbf{x}, i) + \hat{p}_j(\text{oc}_j^{ijk}(\mathbf{x}))B^{2D}(\mathbf{x}, j) + \hat{p}_k(\text{oc}_k^{ijk}(\mathbf{x}))B^{2D}(\mathbf{x}, k) \end{aligned} \quad (\text{Supp.7})$$

Abusing notation, we will drop the superscripts on the terms of the form $(\text{oc}_i^{ijk}(\mathbf{x}))$.

Blended Chart Surfaces are equivariant to translation of the coarse geometry if and only if

$$\hat{c}_{ijk}(\mathbf{x}) = c_{ijk}(\mathbf{x}) + T \text{ for } \mathbf{x} \in E.$$

$\Leftrightarrow$

$$\begin{aligned} & (p_i(\text{oc}_i(\mathbf{x})) + T)B^{2D}(\mathbf{x}, i) + (p_j(\text{oc}_j(\mathbf{x})) + T)B^{2D}(\mathbf{x}, j) + \\ & (p_k(\text{oc}_k(\mathbf{x})) + T)B^{2D}(\mathbf{x}, k) = \\ & p_i(\text{oc}_i(\mathbf{x}))B^{2D}(\mathbf{x}, i) + p_j(\text{oc}_j(\mathbf{x}))B^{2D}(\mathbf{x}, j) + \\ & p_k(\text{oc}_k(\mathbf{x}))B^{2D}(\mathbf{x}, k) + T, \text{ for } \mathbf{x} \in E \end{aligned}$$

$\Leftrightarrow$

$$(B^{2D}(\mathbf{x}, i) + B^{2D}(\mathbf{x}, j) + B^{2D}(\mathbf{x}, k))T = T, \quad \text{for } \mathbf{x} \in E,$$

by subtracting the common terms on both sides.

Using the assumption that $T \neq 0$, we see that the equation holds if and only if

$$B^{2D}(\mathbf{x}, i) + B^{2D}(\mathbf{x}, j) + B^{2D}(\mathbf{x}, k) = 1, \quad \text{for } \mathbf{x} \in E.$$

This is exactly the PoU property. So, equivariance to translation holds if and only if the blending function has PoU.

B.3 The 1D Blending Function is C^k continuous

We want to show that the ‘smooth transition’ blending function B , as defined in Section 3.1, is smooth (i.e., C^k continuous for all k).

Reminder:

$$B(x) = g\left(\frac{|x| - (1 - \beta)/2}{\beta}\right) \quad (\text{Supp.8})$$

where

$$g(x) = \frac{f(1 - x)}{f(x) + f(1 - x)} \quad (\text{Supp.9})$$

and

$$f(x) = \begin{cases} \exp(-\frac{1}{x}) & \text{for } x > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (\text{Supp.10})$$

We first show that f is smooth, and then argue that this implies g and B are smooth. The function g is known as a Smooth Transition Function because it transitions between the functions that are constantly one, and constantly zero, and it has C^∞ smoothness.

Showing that f is smooth:

Clearly, $\frac{d}{dx} f(x) = 0$ for $x < 0$, for all k . Applying the chain rule and product rule, we see that for $x > 0$, every derivative has the form

$$\frac{d}{dx^k} f(x) = P_k(x) e^{-x^{-1}} \quad (\text{Supp.11})$$

where $P_0(x) = 1$, $P_1(x) = x^{-2}$ and in general,

$$P_{k+1}(x) = P'_k(x) + x^{-2} P_k(x) \quad (\text{Supp.12})$$

Without actually solving the recurrence, we see (inductively) that $P_k(x)$ always takes the form

$$P_k(x) = a_{1,k} x^{-(k+1)} + \dots + a_{k,k} x^{-2k} \quad (\text{Supp.13})$$

for some scalar coefficients $a_{i,j}$.

Let $t = 1/x$. Then $f_k(x) = f_k(1/t) = P_k(t) e^{-t}$. Now $P_k(1/t)$ is a polynomial in t , and by Algebra of Limits, $f_k(1/t) \rightarrow 0$ as $t \rightarrow \infty$. Therefore $f_k(x) \rightarrow 0$ as $x \downarrow 0$.

So the right hand and left hand limits agree, therefore f is smooth including at zero.

A similar proof of the smoothness of the function f is given by [1].

Showing that g is smooth:

Since $f(x)$ is smooth, then $f(1 - x)$ is smooth, and $f(x) + f(1 - x)$ is smooth. We also know that $f(x) + f(1 - x)$ is strictly positive for all x , so $\frac{1}{f(x) + f(1 - x)}$ is smooth. Therefore, $g(x)$ is the product of two smooth functions of x , so $g(x)$ is smooth.

Showing that B is smooth:

The parameter β is constrained to be greater than zero. Then $B_+(x) = g\left(\frac{x - (1 - \beta)/2}{\beta}\right)$ and $B_-(x) = g\left(\frac{-x - (1 - \beta)/2}{\beta}\right)$ are clearly smooth because they are simple transformations of $g(x)$.

It remains to show that the left hand and right hand limits agree. That is trivially true, because B_+ and B_- are both constant (equal to 1) in a neighborhood of zero (because we set β strictly less than 1), so all derivatives on both side are zero.

Therefore B has C^∞ continuity, as required.

B.4 The Blended Chart Curve is C^k Continuous

For this part it is cleanest to extend B and the p_i to functions that are zero outside of $[-1, 1]$. Then we can just write

$$c_i(t) = p_i(t)B(t) + p_{i-1}(t+1)B(t+1) + p_{i+1}(t-1)B(t-1)$$

for $t \in [-1, 1]$ and we remove the need for cases in the definition.

The General Leibniz Rule states that if f and g are both n -times differentiable, then their product fg is n -times differentiable (see [2], p. 318). The n^{th} derivative may be expressed using binomial coefficients, but we only require the existence statement.

Now, all of the p_i functions are k -times differentiable by assumption, so $p_i(t)$, $p_{i-1}(t+1)$ and $p_{i+1}(t-1)$ are all k -times differentiable (using the chain rule), and we showed above that $B(t)$ is k -times differentiable for all k , therefore $B(t-1)$ and $B(t+1)$ are also k -times differentiable. Importantly, B is zero at ± 1 and the first k derivatives of B are also zero at ± 1 (Properties C2 and C3). This ensures that the products between local maps p_i and blending functions B are also C^k on the *extended* domain, so smoothness is not broken at the vertex locations (i.e. when $t = 0$ or $t = \pm 1$).

All of the terms $p_i(t)B(t)$, $p_{i-1}(t+1)B(t+1)$ and $p_{i+1}(t-1)B(t-1)$ are k -times differentiable, and so is their sum $c_i(t)$ (by linearity of differentiation).

So c_i is C^k continuous. i.e. c_i inherits the same level of smoothness as the p_i functions, and as long as we construct the p_i functions to be C^∞ continuous then the c_i is C^∞ continuous.

B.5 Allowable range of α , for Blended Chart Surfaces

To ensure that each blending function reaches zero before reaching the opposite edge, α must be smaller than the distance to the opposite edge. (If the blending function does not reach zero before reaching the opposite edge, then there is a jump when the induced ‘hat function’ drops to zero outside of the one-ring, and this leads to discontinuities/ holes in the Blended Chart Surface.) Since the domain triangle E is equilateral with unit side-lengths, this means that $\alpha < \cos(\pi/6) = \sqrt{3}/2$. (See Figure Supp.1, left.)

Additionally, in order for the sum of the blending functions to always be positive (so that it’s possible to normalize to get PoU blending functions), we require that α is greater than the distance to the centre of the triangle. That is, $\alpha > \sqrt{3}/3$. (See Figure Supp.1, right.) Overall, the allowable range of the ‘overlap parameter’ α is $\sqrt{3}/3 < \alpha < \sqrt{3}/2$, or approximately, $0.577 < \alpha < 0.866$.

B.6 The one-ring coordinates map and the transition maps are C^∞ diffeomorphisms

The ‘flattening’ map that converts equilateral-triangle coordinates to one-ring coordinates is a C^∞ diffeomorphism on the punctured one-ring. This is because, except at the origin, it can be expressed in polar coordinates as a smooth composition of three maps: the coordinate change $(x, y) \mapsto (r, \theta)$, the linear angular rescaling $(r, \theta) \mapsto (r, \frac{6\theta}{\eta})$, and the inverse coordinate change $(r, \phi) \mapsto (r \cos \phi, r \sin \phi)$. The composition of smooth maps is smooth. The Jacobian determinant is constant and equal to $\frac{6}{\eta}$ everywhere (which is non-zero); geometrically, the map scales the angular direction by $\frac{6}{\eta}$ while leaving the radial direction unchanged. Therefore the ‘flattening’ map is a C^∞ diffeomorphism.

The composition of an ‘unflattening’ and ‘flattening’ map is still a C^∞ diffeomorphism as long as none of the points lies on one of the ‘centres of flattening’ (because diffeomorphisms are closed under composition and inversion). Therefore the *transition maps* that convert between one-ring coordinates centred on two different vertices, for points in the interior of the rhomboidal overlap region between two one-rings, are C^∞ diffeomorphisms.

B.7 The 2D Blending Function has C^∞ continuity

The 2D blending function is, by construction, 1 at one vertex in the equilateral parametrization domain and then drops outward from the vertex, before reaching zero at the radius α . With a value of α in the valid range, then the radially symmetric heightfield swept out by B is smooth on the equilateral triangle domain (smoothness is ensured at the central vertex, because our choice of the parameter α ensures that B is locally flat in a finite neighborhood of both one and zero).

A corollary of the General Leibniz Rule states that if a and b are both n -times differentiable, and b is nowhere zero, then a/b is n -times differentiable. (This follows because a/b can be expressed as the

product $(a)(1/b)$, and $1/b$ is C^n because the reciprocal function is C^∞ except at zero, and the composition of C^n functions is also C^n .)

Applying this to the definition of B^{2D} , the numerator and denominator are both C^∞ and since we took α to be greater than $\sqrt{3}/3$, then the denominator is nowhere zero, therefore B^{2D} is C^∞ . In other words, when we normalize to ensure PoU, this doesn't break the smoothness of the 2D blending function.

We have shown that B^{2D} is C^∞ with respect to the equilateral triangle domain. In the next section we will need to use the fact that B^{2D} is C^∞ with respect to the *one-ring coordinates*.

Showing that B^{2D} is C^∞ with respect to one-ring coordinates in a single triangle:

The flattening map (from equilateral one-rings to flat one-ring coordinates) is a C^∞ diffeomorphism, everywhere except the central vertex (refer to section C.6). Therefore B^{2D} is C^∞ with respect to one-ring coordinates in a single triangle, except possibly at the central vertex. In fact, B^{2D} is also C^∞ at the central vertex, because our construction ensures that B^{2D} is locally flat, equal to one, in a neighborhood of the vertex (this follows from properties S2 and S3), so the derivatives of all degrees are zero there.

Showing that B^{2D} is C^∞ with respect to one-ring coordinates in the whole one-ring:

It isn't immediately obvious that the 'hat function' formed by tiling together the blending functions on the η triangles into a heightfield on a whole one-ring, is still C^∞ along the edges (with respect to one-ring coordinates).

This 'hat function' is C^∞ with respect to one-ring coordinates though, because B^{2D} was constructed to have all directional derivatives equal to zero along the direction perpendicular to the edges, at points on the edges (property S3), so there is no discontinuity in the gradient or any of the higher derivatives as we cross over an edge.

B.8 The Blended Chart Surface has C^∞ Continuity

We would like to show that the Blended Chart Surface has C^∞ smoothness. To be more precise, we want to show that the Blended Chart Surface has

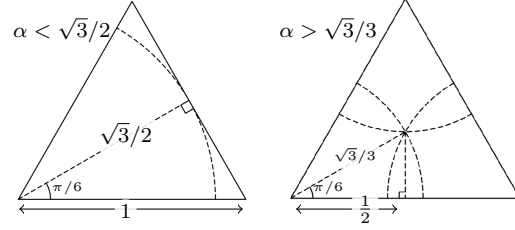


Figure Supp.1: Allowable range of α parameter (the radius of the blending function's non-zero region). Left: $\alpha < \cos(\pi/6) = \sqrt{3}/2$, so that the blending function reaches zero *before* it reaches the opposite edge. Right: $\alpha > \frac{1}{2}/\cos(\pi/6) = \sqrt{3}/3$, so that there is a non-zero amount of overlap in the support of the blending functions from each vertex.

C^∞ smoothness with respect to some overlapping local charts. In particular, we take the local charts to be the one-ring coordinates on the interior of the one-rings (slightly inside the edge of the one-rings, to avoid the outer vertices).

Similar to the proof in the curves case, we can extend the blending functions with zeros outside of the one-ring, to allow us to express the overlapping local-maps c_i without the need for cases:

$$c_i(\text{oc}_i(\mathbf{x})) = B^{2D}(\mathbf{x}, i)p_i(\text{oc}_i(\mathbf{x})) + \sum_{j:(i,j) \in \text{edges}} B^{2D}(\mathbf{x}, j)p_j(\text{oc}_j(\mathbf{x})) \quad (\text{Supp.14})$$

Each of the terms has C^∞ continuity with respect to its own one-ring coordinates; this follows from the C^∞ smoothness of the blending function with respect to one-ring coordinates (section C.7), and the C^∞ smoothness of the p_i functions (which are rigidly transformed polynomials, in our examples).

In addition, the *transition maps* that convert between pairs of one-ring coordinates, e.g. $\text{oc}_i(\mathbf{x})$ and $\text{oc}_j(\mathbf{x})$, are C^∞ diffeomorphisms, away from the vertices i and j (as shown in section C.6).

This means that all of the terms are also C^∞ smooth with respect to the one-ring coordinates of vertex i . Finally, the class of smooth functions is closed under addition, so c_i is C^∞ smooth with respect to the one-ring coordinates of vertex i .

B.9 Can c_i be non-injective, or have a zero-derivative?

Unlike a genuine Atlas, which consists of ‘charts’ from regions of the shape to $\mathbb{R}^d$, we work with *local maps* the other way around: from regions of $\mathbb{R}^d$ to the shape.

This construction does not necessarily constrain c_i to being injective. So, even though c_i is C^k continuous, the image of the curve may contain:

- self-intersections
- singularities where $\frac{d}{dt}c_i(t) = 0$.

Curves that self-intersect are non-manifold. An example would be the path of a particle that moves smoothly and visits the same point at multiple points in time. Singularities may lead to ‘cusps’ in the trace of the curve. (Imagine the path of a particle that slows down to velocity zero, then changes direction when it resumes movement.) For example, if we specifically initialize the vertex functions to be constant points then there are singularities, which show up as sharp corners on the trace of the initialized blended chart curve, but these corners disappear as soon as the p_i functions are perturbed away from zero.

Cusps are undesirable in our curve representation, but this is not practically an issue, because (although we don’t prove this) the probability that the components of a c_i function have zero-derivative at exactly the same point, simultaneously in x and y directions, is zero so this will never actually happen after the initialization.

In the case of Blended Chart Surfaces, self-intersections may also occur when c_i is non-injective. If c_i is singular (some directional derivative is zero) then there can be ‘cusp’-like places, where the surface still has a local parametrisation but the parametrisation ‘slows down to zero’. For example, we see sharp edges if we set all of the vertex functions to be constant points (see Figure 10, far left). But in practice, the probability of the Jacobian being singular after initialisation is zero.

B.10 Is there unique set of vertex functions for a given parametrized shape?

We discuss in the curves case. The question is whether there is a one-to-one correspondence between ordered lists of vertex functions and parametrized curves. In other words, given all of the c_i and knowledge of the blending function, hyperparameters $\alpha, s_{\text{global}}$ etc., can we recover all of the p_i ?

When the p_i belong to a very large class of functions, one can imagine perturbing the three vertex functions for a triangle in such a way that the final blended function, c_i , looks the same or similar. If the class of functions is much smaller, such as polynomials, we could take sample points of the c_i and construct a linear system to solve for the polynomial coefficients. This may be an interesting direction for future work and has relevance to generative models (where it is advantageous for the latent space of ‘shape parameters’ to be compact).

References

- [1] Jet Nestruev. *Smooth Manifolds and Observables*, volume 220 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 2003.
- [2] P.J. Olver. *Applications of Lie Groups to Differential Equations*. Graduate Texts in Mathematics. Springer New York, 1993.